



Sesi 1: Hello Flutter with Nylo



Overview Sesi

Durasi: 90 menit (1.5 jam) | **Branch:** 00-setup-complete → 01-first-run → 02-hello-modified | **Tujuan:** Peserta berhasil menjalankan aplikasi Flutter Nylo pertama mereka dan melakukan modifikasi sederhana



Agenda Sesi

Bagian 1: Setup & Instalasi (30 menit)

- ✓ Install Flutter SDK
- ✓ Install Nylo CLI
- ✓ Verify installation
- ✓ Troubleshooting common issues

Bagian 2: Create First Project (30 menit)

- ✓ Create Nylo project
- ✓ Explore folder structure
- ✓ Run app di emulator/device
- ✓ Hot reload demo

Bagian 3: First Modification (30 menit)

- ✓ Edit HomePage text
 - ✓ Change colors
 - ✓ Add icon dan button
 - ✓ Understand widget tree
-



Bagian 1: Setup & Instalasi (30 menit)

1.1 Install Flutter SDK

Windows:

1. Download Flutter SDK dari <https://docs.flutter.dev/get-started/install/windows>
2. Extract zip file ke lokasi (contoh: C:\src\flutter)
3. Tambahkan Flutter ke PATH:

- Search “Environment Variables” di Windows
- Edit “Path” variable
- Add C:\src\flutter\bin

macOS:

```
# Download Flutter SDK
cd ~/development
git clone https://github.com/flutter/flutter.git -b stable

# Add to PATH (tambahkan ke ~/.zshrc atau ~/.bashrc)
export PATH="$PATH:$HOME/development/flutter/bin"

# Apply changes
source ~/.zshrc
```

Linux:

```
# Download Flutter SDK
cd ~/development
git clone https://github.com/flutter/flutter.git -b stable

# Add to PATH
export PATH="$PATH:$HOME/development/flutter/bin"

# Apply changes
source ~/.bashrc
```

Verify Installation:

```
flutter --version
# Output harus menunjukkan Flutter 3.24.0 atau lebih tinggi

flutter doctor
# Check semua requirements
```

Output yang Diharapkan:

```
Flutter 3.24.0 • channel stable
Tools • Dart 3.5.0 • DevTools 2.37.0

[✓] Flutter (Channel stable, 3.24.0, on macOS)
[✓] Android toolchain - develop for Android devices
[✓] Xcode - develop for iOS and macOS (Xcode 15.0)
[✓] Chrome - develop for the web
[✓] VS Code (version 1.85.0)
[✓] Connected device (2 available)
```

1.2 Understand Metro CLI

Apa itu Metro? Metro adalah CLI tool dari Nylo untuk generate code (models, controllers, pages, dll).

Cara Menggunakan Metro:

Command lengkap (tanpa alias)

```
dart run nylo_framework:main
```

Contoh: Create model

```
dart run nylo_framework:main make:model Task
```

Contoh: Create page

```
dart run nylo_framework:main make:page home_page
```

💡 Optional: Buat Alias untuk Kemudahan

macOS/Linux:

Tambahkan alias ke bash_profile atau zshrc

```
sudo echo "alias metro='dart run nylo_framework:main'" >> ~/.zshrc &&  
source ~/.zshrc
```

Test alias

```
metro
```

Akan muncul Metro CLI menu

Windows (PowerShell):

Buat/edit PowerShell profile

```
if (!(Test-Path -Path $PROFILE)) {  
    New-Item -ItemType File -Path $PROFILE -Force  
}
```

Edit profile

```
notepad $PROFILE
```

Tambahkan line ini:

```
function metro { dart run nylo_framework:main @args }
```

Save dan reload

```
. $PROFILE
```

Setelah setup alias, Anda bisa gunakan metro instead of `dart run nylo_framework:main`.

⚠️ **Catatan:** Metro **tidak perlu** diinstall terpisah. Dia sudah include dalam Nylo project. Anda hanya perlu run `dart run nylo_framework:main` dari dalam project folder.—

1.3 Setup Android Emulator (Optional)

Jika belum punya physical device:

Check available emulators

```
flutter emulators
```

Launch emulator

```
flutter emulators --launch <emulator_id>
```

```
# Atau buat emulator baru via Android Studio:  
# Tools → Device Manager → Create Device → Pilih Pixel 5 → Finish
```

1.4 Verify Everything Works

```
# Check Flutter doctor  
flutter doctor -v
```

```
# List connected devices  
flutter devices
```

```
# Harus menunjukkan minimal 1 device (emulator atau physical)
```

✅ **Checklist Setup:** - [] Flutter SDK installed - [] flutter doctor tidak ada [!] error
- [] Minimal 1 device available (flutter devices) - [] Understand cara akses Metro CLI (dart run nylo_framework:main)



Bagian 2: Create First Project (30 menit)

2.1 Download Nylo Project Template

Ada 2 cara download Nylo:

Option 1: Git Clone (Recommended)

```
# Navigate ke folder workspace  
cd ~/Documents/workshop-flutter
```

```
# Clone Nylo template dari GitHub  
git clone https://github.com/nylo-core/nylo.git my_todo_app
```

```
# Masuk ke folder project  
cd my_todo_app
```

```
# Install dependencies  
flutter pub get
```

Option 2: Download ZIP

1. Buka <https://nylo.dev/download>
2. Download Nylo template (ZIP file)
3. Extract ke folder my_todo_app
4. Buka terminal di folder tersebut
5. Run flutter pub get

🔍 **Apa yang Didapat:** - Flutter project dengan struktur Nylo - Dependencies sudah dikonfigurasi - Boilerplate code siap pakai - Metro CLI tools sudah tersedia—

2.2 Explore Folder Structure

```
cd my_todo_app
```

Struktur Folder:

```
my_todo_app/
├── lib/
│   ├── app/
│   │   ├── controllers/    # Business logic
│   │   ├── models/        # Data structures
│   │   └── networking/     # API calls
│   ├── resources/
│   │   ├── pages/         # UI screens
│   │   │   └── home_page.dart
│   │   └── widgets/       # Reusable widgets
│   ├── routes/
│   │   └── routes.dart    # Route definitions
│   ├── bootstrap/
│   │   └── app.dart       # App initialization
│   └── main.dart          # Entry point
├── pubspec.yaml           # Dependencies
├── .env                   # Environment variables
└── README.md
```

Penjelasan Singkat:

- **lib/app/** → Logic aplikasi (controllers, models)
 - **lib/resources/pages/** → Halaman-halaman UI
 - **lib/routes/** → Navigasi antar halaman
 - **lib/main.dart** → Starting point aplikasi
 - **pubspec.yaml** → Daftar package/library yang digunakan
-

2.3 Run Default App

```
# Install dependencies
flutter pub get
```

```
# Run app
flutter run
```

```
# Atau jika ada multiple devices:
flutter run -d <device_id>
```

Expected Output:

App akan terbuka di emulator/device dengan tampilan default Nylo: - App bar dengan judul “Home” - Body dengan text “Hello World” atau Nylo logo - FloatingActionButton di kanan bawah

 **Congratulations!** Anda sudah menjalankan aplikasi Flutter Nylo pertama!

2.4 Hot Reload Demo

Apa itu Hot Reload? Hot reload memungkinkan Anda melihat perubahan code secara instant tanpa restart app.

Test Hot Reload:

1. **Buka** lib/resources/pages/home_page.dart
2. **Edit** text di dalam widget:

```
// Find this:  
Text('Hello World')
```

```
// Change to:  
Text('Hello Flutter!')
```

3. **Save file** (Cmd+S / Ctrl+S)
4. **Lihat emulator** → Text akan berubah otomatis! ⚡

Shortcut Hot Reload: - **VS Code:** Cmd+S / Ctrl+S (auto hot reload saat save) -

Terminal: Tekan r di terminal yang menjalankan flutter run - **Hot Restart:** Tekan R (capital R) untuk full restart

Bagian 3: First Modification (30 menit)

3.1 Edit HomePage Text

Mari kita ubah tampilan HomePage dengan sesuatu yang lebih personal.

File: lib/resources/pages/home_page.dart

Sebelum:

```
import 'package:flutter/material.dart';  
import 'package:nylo_framework/nylo_framework.dart';  
  
class HomePage extends NyStatefulWidget {  
  static const path = '/home';  
  
  HomePage({Key? key}) : super(key: key);  
  
  @override  
  _HomePageState createState() => _HomePageState();  
}  
  
class _HomePageState extends NyState<HomePage> {  
  @override  
  Widget build(BuildContext context) {  
    return Scaffold(  
      appBar: AppBar(  
        title: Text("Home"),  
      ),  
      body: Center(  

```

```

        child: Text("Hello World"),
      ),
    );
  }
}

```

Setelah:

```

import 'package:flutter/material.dart';
import 'package:nylo_framework/nylo_framework.dart';

class HomePage extends NyStatefulWidget {
  static const path = '/home';

  HomePage({Key? key}) : super(key: key);

  @override
  _HomePageState createState() => _HomePageState();
}

class _HomePageState extends NyState<HomePage> {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("My ToDo App"),
        backgroundColor: Colors.blue,
      ),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Icon(
              Icons.check_circle,
              size: 100,
              color: Colors.blue,
            ),
            SizedBox(height: 20),
            Text(
              "Welcome to My ToDo App!",
              style: TextStyle(
                fontSize: 24,
                fontWeight: FontWeight.bold,
              ),
            ),
            SizedBox(height: 10),
            Text(
              "Let's get things done together",
              style: TextStyle(
                fontSize: 16,
                color: Colors.grey,
              ),
            ),
          ],
        ),
      ),
    ),
  ],
)

```

```

    ),
  ),
);
}
}

```

🎨 **Perubahan yang Dilakukan:** 1. ✅ AppBar title → “My ToDo App” 2. ✅ Added blue background color to AppBar 3. ✅ Added check_circle icon (size 100, blue) 4. ✅ Changed text dengan styling 5. ✅ Used Column untuk vertical layout 6. ✅ Added spacing dengan SizedBox

Save dan lihat hasilnya! 🚀

3.2 Add Button

Sekarang tambahkan button di bawah text.

Update bagian children di Column:

```

Column(
  mainAxisAlignment: MainAxisAlignment.center,
  children: [
    Icon(
      Icons.check_circle,
      size: 100,
      color: Colors.blue,
    ),
    SizedBox(height: 20),
    Text(
      "Welcome to My ToDo App!",
      style: TextStyle(
        fontSize: 24,
        fontWeight: FontWeight.bold,
      ),
    ),
    SizedBox(height: 10),
    Text(
      "Let's get things done together",
      style: TextStyle(
        fontSize: 16,
        color: Colors.grey,
      ),
    ),
    SizedBox(height: 30),
    // 🆕 Button baru
    ElevatedButton(
      onPressed: () {
        print("Button clicked!");
      },
      style: ElevatedButton.styleFrom(
        backgroundColor: Colors.blue,
        padding: EdgeInsets.symmetric(horizontal: 40, vertical: 15),
      ),
    ),
  ],
)

```



```

        child: Text(
          "Get Started",
          style: TextStyle(fontSize: 18),
        ),
      ),
    ],
  ),
)

```

Test Button: 1. Save file 2. Tap button di app 3. Lihat console → akan muncul “Button clicked!”

3.3 Understand Widget Tree

Apa itu Widget Tree?

Di Flutter, semua yang Anda lihat di screen adalah widget. Widget disusun dalam tree structure.

Contoh Widget Tree untuk HomePage kita:

```

Scaffold
├── AppBar
├── Text ("My ToDo App")
├── body: Center
│   └── Column
│       ├── Icon (check_circle)
│       ├── SizedBox (spacing)
│       ├── Text ("Welcome...")
│       ├── SizedBox (spacing)
│       ├── Text ("Let's get things done...")
│       ├── SizedBox (spacing)
│       └── ElevatedButton
│           └── Text ("Get Started")

```

💡 Key Concepts:

- **Scaffold:** Blueprint halaman (AppBar, body, FloatingActionButton)
 - **Center:** Widget yang center childnya
 - **Column:** Layout vertical (dari atas ke bawah)
 - **Row:** Layout horizontal (dari kiri ke kanan)
 - **SizedBox:** Spacing atau container dengan size tertentu
 - **Padding:** Add spacing around widget
-

3.4 Experiment Time! ✏️

Coba modifikasi sendiri:

Challenge 1: Change Colors

Ubah warna icon dan button ke hijau (Colors.green)

Challenge 2: Add More Text

Tambahkan text di bawah button: "Version 1.0"

Challenge 3: Add Padding

Wrap Column dengan Padding widget:

```
body: Center(  
  child: Padding(  
    padding: EdgeInsets.all(20),  
    child: Column(  
      // ... existing children  
    ),  
  ),  
)
```

Challenge 4: Add FloatingActionButton

Tambahkan FloatingActionButton di Scaffold:

```
return Scaffold(  
  appBar: AppBar(...),  
  body: Center(...),  
  floatingActionButton: FloatingActionButton(  
    onPressed: () {  
      print("FAB clicked!");  
    },  
    child: Icon(Icons.add),  
    backgroundColor: Colors.blue,  
  ),  
);
```

Testing & Verification

Checklist Sesi 1:

- ☐ Flutter SDK installed dan verified
 - ☐ Nylo CLI installed
 - ☐ Project created dengan `nylo create`
 - ☐ App berjalan di device/emulator
 - ☐ Hot reload works
 - ☐ HomePage modified (text, icon, button)
 - ☐ Button onClick works (print di console)
 - ☐ Understand basic widget structure
-



Common Issues & Solutions

Issue 1: “Command ‘flutter’ not found”

Symptom: Terminal tidak recognize flutter command

Solution:

```
# Verify Flutter location
which flutter

# If not found, add to PATH
export PATH="$PATH:/path/to/flutter/bin"
source ~/.zshrc
```

Issue 2: “Android licenses not accepted”

Symptom: flutter doctor shows Android licenses issue

Solution:

```
flutter doctor --android-licenses
# Press 'y' untuk accept semua licenses
```

Issue 3: “No devices found”

Symptom: flutter devices returns empty

Solution:

For Physical Device: - Enable USB Debugging di Developer Options - Connect via USB
- Trust computer saat diminta

For Emulator:

```
# List available emulators
flutter emulators

# Launch specific emulator
flutter emulators --launch Pixel_5_API_33
```

Issue 4: “Gradle build failed”

Symptom: Error saat build Android app

Solution:

```
# Clean build
flutter clean

# Get dependencies
```

```
flutter pub get
```

```
# Run again
```

```
flutter run
```

Issue 5: “Hot reload not working”

Symptom: Perubahan code tidak muncul di app

Solution: - Press R (capital R) untuk hot restart - Atau stop dan run ulang app - Check if you saved the file (Cmd+S / Ctrl+S)



Key Takeaways

✅ **Flutter SDK:** - Cross-platform framework (Android, iOS, Web, Desktop) - Uses Dart programming language - Hot reload untuk fast development - Widget-based UI architecture

✅ **Nylo Framework:** - Micro framework built on top of Flutter - Provides MVC-like structure - Metro CLI untuk scaffolding (make:model, make:page, dll) - Built-in routing dan state management

✅ **Widget Basics:** - Everything is a widget - Widgets arranged in tree structure - Stateless vs Stateful widgets - Common widgets: Scaffold, AppBar, Text, Icon, Button

✅ **Development Workflow:** - Edit code → Save → Hot reload → See changes - Use `print()` untuk debugging - Check console untuk errors dan logs



Resources

Official Docs:

- Flutter: <https://docs.flutter.dev>
- Nylo: <https://nylo.dev/docs/6.x>
- Dart: <https://dart.dev>

Tutorials:

- Flutter Codelabs: <https://docs.flutter.dev/codelabs>
- Nylo Quick Start: <https://nylo.dev/docs/6.x/installation>
- Flutter Widget Catalog: <https://docs.flutter.dev/development/ui/widgets>

Videos:

- Flutter Official YouTube: <https://youtube.com/flutter>
- Flutter Widget of the Week: <https://youtube.com/playlist?list=PLjxrf2q8roU23XGwz3Km7sQZFTdB996iG>

 Navigation

[Workshop Modules](#) | [Sesi 2 - UI & Navigation](#) →

Workshop Material - Simple ToDo List with Flutter Nylo | Dokumentasi Sesi 1 - Hello Flutter with Nylo | Terakhir diperbarui: November 21, 2025